1. What is the forbidden energy bandgap of Silicon at room temperature (300 K)?

(A) 1.1 eV
(B) 0.7 eV
(C) 0.0 eV
(D) 5.0 eV

2. What is the forbidden energy bandgap of Germanium at room temperature?

(A) 0.72 eV
(B) 1.1 eV
(C) 1.42 eV
(D) 0.1 eV

3. A pentavalent impurity added to intrinsic silicon creates an:

(A) N-type semiconductor
(B) P-type semiconductor
(C) Insulator
(D) Intrinsic semiconductor

4. A trivalent impurity (such as Boron or Gallium) added to silicon creates a:

(A) P-type semiconductor
(B) N-type semiconductor
(C) Metal
(D) Superconductor

5. In an N-type semiconductor, the majority charge carriers are:

(A) Electrons
(B) Holes
(C) Positive ions
(D) Photons

6. In a P-type semiconductor, the majority charge carriers are:

(A) Holes
(B) Electrons
(C) Negative ions
(D) Neutrons

7. The barrier potential (knee voltage) of a silicon PN junction diode at 300 K is approximately:

(A) 0.7 V
(B) 0.3 V
(C) 1.1 V
(D) 0.1 V

8. The barrier potential of a germanium PN junction diode is approximately:

(A) 0.3 V
(B) 0.7 V
(C) 1.2 V
(D) 0.0 V

9. In a forward-biased PN junction, the depletion layer width:

(A) Decreases
(B) Increases
(C) Remains constant
(D) Becomes infinite

10. In a reverse-biased PN junction, the depletion layer width:

(A) Increases
(B) Decreases
(C) Disappears
(D) Remains unchanged

11. Reverse saturation current in a silicon diode approximately doubles for every temperature rise of:

(A) 10 deg C
(B) 5 deg C
(C) 20 deg C
(D) 1 deg C

12. The ripple factor of an unfiltered Half-Wave Rectifier is:

(A) 1.21
(B) 0.482
(C) 1.11
(D) 0.707

13. The ripple factor of an unfiltered Full-Wave Bridge Rectifier is:

(A) 0.482
(B) 1.21
(C) 0.1
(D) 1.414

14. The maximum theoretical rectification efficiency of a Half-Wave Rectifier is:

(A) 40.6%
(B) 81.2%
(C) 50.0%
(D) 100%

15. The maximum theoretical rectification efficiency of a Full-Wave Rectifier is:

(A) 81.2%
(B) 40.6%
(C) 90.0%
(D) 100%

16. The output ripple frequency of a half-wave rectifier fed by a 50 Hz AC supply is:

(A) 50 Hz
(B) 100 Hz
(C) 25 Hz
(D) 0 Hz

17. The output ripple frequency of a full-wave rectifier fed by a 50 Hz AC supply is:

(A) 100 Hz
(B) 50 Hz
(C) 200 Hz
(D) 25 Hz

18. How many diodes are required to construct a Full-Wave Bridge Rectifier?

(A) 4
(B) 2
(C) 1
(D) 6

19. Peak Inverse Voltage (PIV) across each diode in a Full-Wave Bridge Rectifier is:

(A) Vm
(B) 2 * Vm
(C) Vm / 2
(D) Vm / sqrt(2)

20. Peak Inverse Voltage (PIV) across each diode in a Center-Tapped Full-Wave Rectifier is:

(A) 2 * Vm
(B) Vm
(C) Vm / 2
(D) 4 * Vm

21. A Zener diode is operated primarily in which biasing region?

(A) Reverse Breakdown region
(B) Forward Active region
(C) Cutoff region
(D) Saturation region

22. A Zener diode is widely used in electronic circuits as a:

(A) Shunt voltage regulator
(B) Current amplifier
(C) Oscillator
(D) Rectifier only

23. In a Bipolar Junction Transistor (BJT), the three terminals are:

(A) Emitter, Base, Collector
(B) Anode, Cathode, Gate
(C) Source, Drain, Gate
(D) Input, Output, Ground

24. Which region of a BJT is the most heavily doped?

(A) Emitter
(B) Base
(C) Collector
(D) Depletion zone

25. Which region of a BJT has the largest physical cross-sectional area to dissipate heat?

(A) Collector
(B) Emitter
(C) Base
(D) Channel

26. Which region of a BJT is lightly doped and extremely thin?

(A) Base
(B) Emitter
(C) Collector
(D) Gate

27. For a BJT to operate in the Active (Amplifying) Region, the junctions must be biased as:

(A) Emitter-Base Forward, Collector-Base Reverse
(B) Both Forward
(C) Both Reverse
(D) Emitter-Base Reverse, Collector-Base Forward

28. When both junctions of a BJT are forward-biased, the transistor operates in:

(A) Saturation region (ON switch)
(B) Cutoff region
(C) Active region
(D) Breakdown region

29. When both junctions of a BJT are reverse-biased, the transistor operates in:

(A) Cutoff region (OFF switch)
(B) Saturation region
(C) Active region
(D) Linear region

30. The common-base current gain alpha is defined as:

(A) Delta-Ic / Delta-Ie
(B) Delta-Ic / Delta-Ib
(C) Delta-Ie / Delta-Ib
(D) Delta-Ib / Delta-Ic

31. The common-emitter current gain beta is related to alpha by:

(A) beta = alpha / (1 - alpha)
(B) beta = alpha / (1 + alpha)
(C) beta = (1 - alpha) / alpha
(D) beta = 1 / alpha

32. If a transistor has alpha = 0.98, its common-emitter current gain beta is:

(A) 49
(B) 50
(C) 98
(D) 0.02

33. Which BJT configuration provides both high voltage gain and high current gain?

(A) Common Emitter (CE)
(B) Common Base (CB)
(C) Common Collector (CC)
(D) Cascode

34. Common Collector (CC) configuration is also known as an:

(A) Emitter Follower
(B) Source Follower
(C) Inverter
(D) Current Mirror

35. The voltage gain of an Emitter Follower is approximately:

(A) Slightly less than unity (~1.0)
(B) 100
(C) 0
(D) Infinite

36. Which logic gates are termed Universal Gates?

(A) NAND and NOR
(B) AND and OR
(C) XOR and XNOR
(D) NOT and AND

37. According to De Morgan's first theorem, (A + B)' is equivalent to:

(A) A' . B'
(B) A' + B'
(C) A . B
(D) (A . B)'

38. According to De Morgan's second theorem, (A . B)' is equivalent to:

(A) A' + B'
(B) A' . B'
(C) A + B
(D) A' . B

39. In Boolean algebra, A + A' is equal to:

(A) 1
(B) 0
(C) A
(D) A'

40. In Boolean algebra, A . A' is equal to:

(A) 0
(B) 1
(C) A
(D) 2A

41. Zener breakdown occurs in PN junctions that are:

(A) Heavily doped and have narrow depletion width (< 6 V)
(B) Lightly doped (> 6 V)
(C) Unbiased
(D) Forward-biased

42. Avalanche breakdown occurs via impact ionization in junctions that are:

(A) Lightly doped with wide depletion region (> 6 V)
(B) Heavily doped (< 4 V)
(C) Forward-biased
(D) Supercooled

43. The temperature coefficient of breakdown voltage in Zener breakdown is:

(A) Negative
(B) Positive
(C) Zero
(D) Infinite

44. The temperature coefficient of breakdown voltage in Avalanche breakdown is:

(A) Positive
(B) Negative
(C) Zero
(D) Exponential

45. A Schottky diode exhibits very fast switching speeds primarily because:

(A) It is a majority-carrier metal-semiconductor junction with zero minority carrier storage
(B) It has high capacitance
(C) It uses germanium
(D) It operates at 1000 V

46. A Varactor diode acts as a voltage-controlled variable:

(A) Capacitor under reverse bias
(B) Resistor
(C) Inductor
(D) Current source

47. Tunnel diodes exhibit negative differential resistance due to:

(A) Quantum mechanical tunneling across ultra-thin barrier
(B) Avalanche multiplication
(C) Thermal runaway
(D) Piezoelectric effect

48. The phase shift between input and output voltages in a Common-Emitter (CE) amplifier is:

(A) 180 degrees
(B) 0 degrees
(C) 90 degrees
(D) 270 degrees

49. The early effect (base-width modulation) in BJTs causes:

(A) Collector current to increase slightly with increasing Vce (finite output resistance ro)
(B) Base width to increase
(C) Beta to drop to zero
(D) Thermal breakdown

50. In BJT biasing, the Stability Factor S is defined as:

(A) d(Ic) / d(Ico)
(B) d(Vbe) / d(T)
(C) d(beta) / d(Ic)
(D) d(Vce) / d(Ib)

51. The ideal value of stability factor S for perfect thermal stability is:

(A) 1.0
(B) Infinity
(C) 0
(D) 100

52. Which BJT biasing circuit provides the most superior thermal stability (lowest S)?

(A) Voltage Divider Bias (Self Bias)
(B) Fixed Bias
(C) Collector-to-Base Bias
(D) Emitter Bias without Re

53. In a Voltage Divider Bias circuit, the stability factor S approaches 1 when:

(A) R_th / R_E is made very small (R_E >> R_th / beta)
(B) R_E is zero
(C) R_th is infinite
(D) Beta is small

54. The DC load line of a CE amplifier connects which two points on the output characteristics?

(A) (Vce = Vcc, Ic = 0) and (Vce = 0, Ic = Vcc / Rc)
(B) (Vcc/2, 0) and (0, Vcc)
(C) (0,0) and (Vcc, Vcc/Rc)
(D) (Vcc, Vcc) and (0,0)

55. Thermal runaway in a BJT occurs when:

(A) Rate of heat generation in collector junction exceeds rate of thermal dissipation
(B) Base current drops
(C) Collector voltage is zero
(D) Load resistor cools down

56. To avoid thermal runaway, the quiescent collector-emitter voltage Vce must satisfy:

(A) Vce < Vcc / 2
(B) Vce > Vcc / 2
(C) Vce = Vcc
(D) Vce = 0

57. The binary equivalent of decimal number 25 is:

(A) 11001
(B) 10101
(C) 11100
(D) 10011

58. The hexadecimal equivalent of binary 1101 1010 is:

(A) DA
(B) CD
(C) BA
(D) EA

59. The 2's complement of 8-bit binary number 0001 0110 (decimal 22) is:

(A) 1110 1010
(B) 1110 1001
(C) 0001 0111
(D) 1111 0000

60. In BCD (8421 code), how many invalid combinations exist in a 4-bit nibble?

(A) 6 (from 1010 to 1111)
(B) 4
(C) 8
(D) 10

61. Gray code is also known as a:

(A) Unit-distance or reflected code
(B) Weighted code
(C) Error-correcting code
(D) Complementary code

62. How many 2-input NAND gates are required to implement a 2-input XOR gate?

(A) 4
(B) 3
(C) 5
(D) 6

63. How many 2-input NOR gates are required to implement a 2-input XOR gate?

(A) 5
(B) 4
(C) 3
(D) 6

64. A Karnaugh Map (K-Map) groups adjacent minterms using:

(A) Gray code adjacency
(B) Binary counting
(C) BCD order
(D) ASCII sequence

65. In a 4-variable K-map, grouping 8 adjacent cells (an octet) eliminates how many variables?

(A) 3 variables
(B) 2 variables
(C) 4 variables
(D) 1 variable

66. In a 4-variable K-map, grouping 4 adjacent cells (a quad) eliminates how many variables?

(A) 2 variables
(B) 3 variables
(C) 1 variable
(D) 4 variables

67. A Half Adder circuit computes the sum of two 1-bit inputs using:

(A) XOR gate for Sum, AND gate for Carry
(B) OR for Sum, AND for Carry
(C) XNOR for Sum, OR for Carry
(D) NAND for both

68. A Full Adder can be constructed using:

(A) Two Half Adders and one OR gate
(B) Two Half Adders only
(C) One Half Adder and one XOR
(D) Three Half Adders

69. A 4-to-1 Multiplexer requires how many select input lines?

(A) 2
(B) 1
(C) 3
(D) 4

70. An 8-to-1 Multiplexer requires how many select lines?

(A) 3
(B) 8
(C) 2
(D) 4

71. A Demultiplexer performs the inverse operation of a Multiplexer and functions as a:

(A) Decoder with an enable input
(B) Encoder
(C) Parity generator
(D) Comparator

72. How many 4-to-1 multiplexers are needed to construct a 16-to-1 multiplexer?

(A) 5 (four in first stage, one in second stage)
(B) 4
(C) 8
(D) 2

73. A 3-to-8 line decoder produces how many active outputs for each input code?

(A) 1
(B) 3
(C) 8
(D) 2

74. A Priority Encoder resolves multiple active inputs by encoding:

(A) The input line with the highest priority
(B) All inputs simultaneously
(C) The lowest priority only
(D) Zero

75. The boolean expression Y = A'B + AB' represents which standard logic gate?

(A) XOR gate
(B) XNOR gate
(C) NAND gate
(D) NOR gate

76. The Einstein relation connecting diffusion coefficient D and carrier mobility mu is:

(A) D / mu = (k * T) / q = Vt
(B) D * mu = Vt
(C) D / mu = q / (k * T)
(D) mu / D = k * T

77. The thermal voltage Vt at room temperature (300 K) is approximately:

(A) 25.86 mV (~26 mV)
(B) 1.0 V
(C) 0.7 V
(D) 100 mV

78. The Shockley ideal diode equation is expressed as:

(A) I = Is * [exp(V / (n * Vt)) - 1]
(B) I = Is * exp(V / Vt)
(C) I = V / Rd
(D) I = Is * [1 - exp(-V/Vt)]

79. Diffusion capacitance (Cd) in a forward-biased diode is proportional to:

(A) Forward diode current IF and carrier lifetime tau
(B) Reverse voltage
(C) Depletion width
(D) Dielectric constant

80. Transition capacitance (Ct) in a reverse-biased diode varies with reverse voltage Vr as:

(A) Ct proportional to 1 / (Vbi + Vr)^m (where m = 0.5 for abrupt junction)
(B) Ct proportional to Vr
(C) Ct is constant
(D) Ct proportional to Vr^2

81. In a full-wave rectifier with capacitor filter, ripple factor is given by:

(A) r = 1 / [4 * sqrt(3) * f * C * RL]
(B) r = 1 / [2 * sqrt(3) * f * C * RL]
(C) r = 2 * sqrt(3) / (f * C)
(D) r = 1 / (f * C)

82. In a BJT, the Ebers-Moll model describes the transistor as:

(A) Two back-to-back interacting ideal diodes with coupled dependent current sources
(B) Two resistors
(C) A pure voltage amplifier
(D) A single PN junction

83. The unity-gain cutoff frequency f_T of a BJT is related to transconductance gm and capacitances by:

(A) f_T = gm / [2 * pi * (C_pi + C_mu)]
(B) f_T = gm * (C_pi + C_mu)
(C) f_T = 1 / [2 * pi * gm * C_pi]
(D) f_T = gm / C_mu

84. In Miller's Theorem, a feedback capacitance Cf connected between input and inverting output with voltage gain -Av appears across input as:

(A) Cin = Cf * (1 + Av)
(B) Cin = Cf * (1 - Av)
(C) Cin = Cf / Av
(D) Cin = Cf

85. The high-frequency roll-off of a CE amplifier is predominantly caused by:

(A) Miller multiplication of collector-base junction capacitance C_mu
(B) Coupling capacitors
(C) Bypass capacitor
(D) Load resistance

86. In digital circuits, Fan-Out is defined as the:

(A) Maximum number of standard logic gate inputs that an output can drive reliably
(B) Power dissipation per gate
(C) Propagation delay
(D) Noise margin

87. Propagation delay (t_pd) of a logic gate is the average of:

(A) (t_pHL + t_pLH) / 2
(B) t_rise + t_fall
(C) t_setup - t_hold
(D) t_clock

88. The Power-Delay Product (PDP) of a logic family is a figure of merit measured in:

(A) Joules (Watt-seconds or picojoules)
(B) Watts
(C) Nanoseconds
(D) Volts

89. Static CMOS gates dissipate almost zero power during quiescent state because:

(A) Either the PMOS network or NMOS network is always completely OFF
(B) They use zero voltage
(C) Resistance is zero
(D) Current is infinite

90. In CMOS logic, dynamic power dissipation is governed by:

(A) P_dyn = C_L * Vdd^2 * f_clk
(B) P_dyn = Vdd * I_static
(C) P_dyn = C_L * Vdd * f_clk
(D) P_dyn = Vdd^2 / R

91. A Look-Ahead Carry Adder (CLA) drastically speeds up addition by calculating carry signals:

(A) In parallel using Carry Generate (Gi = Ai*Bi) and Carry Propagate (Pi = Ai^Bi)
(B) Using serial ripple propagation
(C) Using shift registers
(D) Without logic gates

92. The carry generate term Gi for stage i in a CLA is:

(A) Gi = Ai . Bi
(B) Gi = Ai + Bi
(C) Gi = Ai ^ Bi
(D) Gi = (Ai . Bi)'

93. The carry propagate term Pi for stage i in a CLA is:

(A) Pi = Ai ^ Bi (or Ai + Bi)
(B) Pi = Ai . Bi
(C) Pi = Ai' . Bi
(D) Pi = Ai

94. A static hazard in a combinational logic circuit occurs due to:

(A) Unequal propagation delays along converging signal paths producing transient glitches
(B) High temperature
(C) Supply noise
(D) Floating inputs

95. Static-1 hazards can be completely eliminated from a sum-of-products K-map implementation by:

(A) Adding redundant consensus product terms that bridge adjacent groupings
(B) Inverting inputs
(C) Reducing clock frequency
(D) Adding buffers

96. A magnitude comparator circuit comparing two 4-bit numbers A and B outputs A > B when:

(A) The most significant differing bit of A is 1 while the corresponding bit of B is 0
(B) A = B
(C) Sum of bits is even
(D) Parity is odd

97. A BCD-to-7-Segment Decoder driving a common-anode LED display requires active:

(A) LOW outputs (zeros illuminate segments)
(B) HIGH outputs
(C) Pulsed outputs
(D) Tri-state outputs

98. The Quine-McCluskey (tabular) method is preferred over Karnaugh mapping when:

(A) Minimizing Boolean functions with 5 or more variables or for automated computer synthesis
(B) Variables are 2 or 3
(C) No computers exist
(D) Only NAND gates are used

99. A ROM (Read-Only Memory) architecture can implement any arbitrary combinational logic of n inputs and m outputs using:

(A) A fixed full-input decoder followed by programmable OR array
(B) Programmable AND and fixed OR
(C) Flip-flops
(D) Multiplexers only

100. A Programmable Logic Array (PLA) differs from a PAL because in a PLA:

(A) Both the AND array and the OR array are fully programmable
(B) Only AND array is programmable
(C) Only OR array is programmable
(D) Neither is programmable

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Derive the built-in potential (contact potential) and depletion layer width of an abrupt PN junction from Poisson's equation.

Q2. Contrast Zener Breakdown and Avalanche Breakdown mechanisms, highlighting doping levels, electric fields, and temperature coefficients.

Q3. Derive the Ripple Factor, Rectification Efficiency, and Peak Inverse Voltage (PIV) for a Full-Wave Bridge Rectifier.

Q4. Analyze the operation of a Zener Diode Shunt Voltage Regulator, deriving the range of source resistor Rs for regulation.

Q5. Explain the physical operation of an NPN Bipolar Junction Transistor in Active Mode, deriving carrier currents.

Q6. Compare Common Base (CB), Common Emitter (CE), and Common Collector (CC) BJT configurations.

Q7. Explain BJT Biasing: Derive the Stability Factor S for Fixed Bias and Voltage Divider Bias circuits.

Q8. Explain Thermal Runaway in a BJT and derive the mathematical condition to prevent it.

Q9. State and prove De Morgan's Theorems using truth tables and Boolean algebraic proofs.

Q10. Prove that NAND and NOR are Universal Logic Gates by implementing NOT, AND, OR, and XOR gates using NAND only.

Q11. Explain Karnaugh Map (K-Map) minimization, solving a 4-variable function with Don't Care conditions.

Q12. Design a Full Adder circuit using basic logic gates, deriving Sum and Carry expressions and truth tables.

Q13. Explain the design and operation of an 8-to-1 Multiplexer and demonstrate function implementation.

Q14. Explain the Carry Look-Ahead Adder (CLA), deriving carry generate and propagate equations.

Q15. Explain Static and Dynamic Hazards in combinational logic, and explain hazard-free circuit design.

Q16. Compare CMOS and TTL logic families regarding propagation delay, power dissipation, fan-out, and noise margins.

Q17. Explain the working principle and characteristics of Light Emitting Diodes (LED) and Photodiodes.

Q18. Explain the operation of a Varactor Diode, deriving the junction capacitance formula under reverse bias.

Q19. Explain the Tunnel Diode: Energy band diagrams, negative differential resistance, and microwave oscillator applications.

Q20. Explain the High-Frequency Hybrid-Pi Model of a BJT and derive the expression for Miller capacitance.

Q21. Design a 2-bit Magnitude Comparator combinational circuit, deriving truth tables and Boolean expressions.

Q22. Explain Priority Encoders, designing a 4-to-2 Priority Encoder with truth table and K-map minimization.

Q23. Explain BCD-to-7-Segment Decoders and display driving circuits (Common Anode and Common Cathode).

Q24. Explain Programmable Logic Devices (PLDs): Contrast PROM, PAL, and PLA architectures.

Q25. Analyze the C-Filter, L-Filter, and Pi-Section Filter for full-wave rectifiers, deriving ripple factor equations.

Q26. Explain the Ebers-Moll Model of a BJT and derive equations for terminal currents across all operating regions.

Q27. Explain Static CMOS Inverter: Voltage Transfer Characteristic (VTC), noise margins, and propagation delay.

Q28. Explain Quine-McCluskey (Tabular) Minimization Method with a step-by-step worked example.

Q29. Explain the Design of a Look-Up Table (LUT) in Field Programmable Gate Arrays (FPGAs).

Q30. Describe Semiconductor Photolithography and the Planar Process used in monolithic IC fabrication.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [A]
Q3: [A]
Q4: [A]
Q5: [A]
Q6: [A]
Q7: [A]
Q8: [A]
Q9: [A]
Q10: [A]
Q11: [A]
Q12: [A]
Q13: [A]
Q14: [A]
Q15: [A]
Q16: [A]
Q17: [A]
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Q89: [A]
Q90: [A]
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Q94: [A]
Q95: [A]
Q96: [A]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

- Newly freed carriers accelerate and dislodge additional carriers, creating an exponential carrier avalanche.
- Breakdown Voltage: Occurs at higher reverse voltages (> 6 V).
- Temperature Coefficient: Positive. Higher temperature enhances lattice vibrations (phonons), increasing carrier scattering and shortening mean free path. Carriers require higher applied voltage to acquire ionizing kinetic energy: dVa / dT > 0.

[illegible]

- Load current: IL = Vz / RL.
- By KCL at output node: Is = Iz + IL => Iz = Is - IL.

3. Design Bounds for Regulation:

- To maintain regulation, Zener diode must remain in breakdown: Iz >= Iz_min (knee current, typically 1-5 mA).
- To prevent thermal destruction, Zener current must not exceed maximum rated current: Iz <= Iz_max = Pz_max / Vz.

4. Determining Limits of Series Resistor Rs:

- Case 1: Minimum Input Voltage Vin_min and Maximum Load Current IL_max:
Zener current is at minimum: Iz_min = Is - IL_max = [(Vin_min - Vz) / Rs_max] - IL_max.
Rs_max = (Vin_min - Vz) / (Iz_min + IL_max).
- Case 2: Maximum Input Voltage Vin_max and Minimum Load Current IL_min:
Zener current is at maximum: Iz_max = [(Vin_max - Vz) / Rs_min] - IL_min.
Rs_min = (Vin_max - Vz) / (Iz_max + IL_min).
- The selected resistor must satisfy: Rs_min <= Rs <= Rs_max.

[illegible]

- Input: Emitter; Output: Collector; Common: Base.
- Input Impedance (Rin): Very Low (~10 - 50 Ohms).
- Output Impedance (Rout): Very High (~1 - 2 MOhms).
- Current Gain: alpha < 1.0 (no current amplification).
- Voltage Gain: High (~100 - 500).
- Phase Shift: 0 degrees (non-inverting).
- Primary Applications: High-frequency RF amplifiers, impedance matching for low-Z sources.

2. Common Emitter (CE):

- Input: Base; Output: Collector; Common: Emitter.
- Input Impedance (Rin): Medium (~1 - 5 kOhms).
- Output Impedance (Rout): Medium-High (~40 - 100 kOhms).
- Current Gain: beta High (~50 - 300).
- Voltage Gain: High (~100 - 500).
- Power Gain: Highest of all configurations.
- Phase Shift: Strictly 180 degrees (inverting).
- Primary Applications: General-purpose audio, RF, and digital switching amplifiers.

3. Common Collector (CC / Emitter Follower):

- Input: Base; Output: Emitter; Common: Collector.
- Input Impedance (Rin): Very High (~100 - 500 kOhms).
- Output Impedance (Rout): Very Low (~10 - 50 Ohms).
- Current Gain: gamma = 1 + beta High.
- Voltage Gain: Slightly less than unity (~0.98 - 0.99).
- Phase Shift: 0 degrees (in phase).
- Primary Applications: Buffer amplifier, impedance matching between high-Z stages and low-Z loads (speakers).

>> Solution to Question 7:

[Problem Statement]: Explain BJT Biasing: Derive the Stability Factor S for Fixed Bias and Voltage Divider Bias circuits.

- Substituting into general stability equation:
S_vdb = (1 + beta) / [1 + beta * (Re / (Rth + Re))] = (1 + beta) * [1 + (Rth / Re)] / [1 + beta + (Rth / Re)].
- When Rth / Re << 1: S approaches 1.0 (Ideal thermal stability!).

>> **Solution to Question 8:** [illegible]

2. De Morgan's Second Theorem:

- Statement: The complement of a logical product (AND) is equal to the logical sum (OR) of the complements:

$(A . B)' = A' + B'$.

- Proof via Truth Table:

A	B	A.B	(A.B)'	A'	B'	A'+B'
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

The columns (A.B)' and A'+B' match identically. Q.E.D.

3. Significance in Digital Design:

- Enables conversion of any AND-OR network into an all-NAND or all-NOR network.
- Forms the mathematical foundation for bubble logic and circuit simplification.

[illegible]

2. K-Map Grid Construction:
- Rows indexed by AB (00, 01, 11, 10); Columns indexed by CD (00, 01, 11, 10) in Gray code sequence.
- Plot minterms '1' at cells 1, 3, 7, 11, 15.
- Plot don't cares 'X' at cells 0, 2, 5.
- Plot zeros '0' at remaining cells.

3. Cell Grouping Strategy:
- Notice column CD = 11: Cells 3, 7, 11, 15 are all 1s. This forms a complete 4-cell column Quad (cells 3, 7, 11, 15).
- Inspect Row AB = 00: Cells 0 (X), 1 (1), 3 (1), 2 (X) form a 4-cell row Quad!
- Can we combine row 00 and column 11 with other cells?

Look at cells (0, 1, 2, 3, 4, 5, 6, 7): Cells 0(X), 1(1), 2(X), 3(1), 5(X), 7(1) together with empty cells.
Actually, consider the 4-cell column CD=11: Cells (3, 7, 11, 15) -> Group 1. Term: C . D.
Consider the entire top row AB = 00 (cells 0, 1, 2, 3): All four cells are either 1 or X!
Grouping cells 0, 1, 2, 3 yields: A' . B'.
Are all 1s covered?
* Cell 1: covered by A'B'
* Cell 3: covered by A'B' and CD
* Cell 7: covered by CD
* Cell 11: covered by CD
* Cell 15: covered by CD
- All minterms {1, 3, 7, 11, 15} are 100% covered!

4. Minimal Sum-of-Products Expression:
F = A'B' + CD.

>> Solution to Question 12:

Problem Statement: Design a Full Adder circuit using basic logic gates, deriving Sum and Carry expressions and truth tables.

- Half Adder 1: S1 = A ^ B; C1 = A . B.
- Half Adder 2: S = S1 ^ Cin = (A ^ B) ^ Cin; C2 = S1 . Cin = (A ^ B) . Cin.
- Output Carry: Cout = C1 + C2 = AB + (A ^ B) . Cin.

>> Solution to Question 14: Carry Look-Ahead Adder (CLA) - derive carry generate and propagate equations.

- Problem Statement: Explain the design and operation of an 8-to-1 Multiplexer and demonstrate function implementation.

- All carries C1, C2, C3, C4 are evaluated in exactly 2 gate delays regardless of bit length!

[illegible] and noise margins.

>> Solution to Question 18 [illegible] oscillator applications.

- Depletion width is extraordinarily narrow (< 10 nm, approx 100 atomic layers).

2. Operating Characteristics and Negative Resistance:

- Zero Bias: Fermi levels align; no net tunneling current.
- Small Forward Bias (0 to Vp ~ 50 mV): Conduction electrons on N-side align with unoccupied states in valence band on P-side. Electrons tunnel directly through the barrier, producing high Peak Current Ip.
- Increasing Forward Bias (Vp to Vv): As bias increases, valence band states drop below conduction band electrons. Available tunneling states decrease; current drops from Peak Ip to Valley current Iv.

* Dynamic resistance: rd = dV / dI < 0 (Negative Differential Resistance region!).

- Beyond Valley Voltage Vv: Conventional forward diffusion current takes over.

3. Applications:

- Ultra-high-speed switching circuits, microwave oscillators, and reflection amplifiers operating up to tens of Gigahertz.

[illegible]

x1 = (A1 ^ B1)' = A1.B1 + A1'.B1'

x0 = (A0 ^ B0)' = A0.B0 + A0'.B0'.

- Output (A = B):

(A = B) = x1 . x0 = (A1 ^ B1)' . (A0 ^ B0)'.

3. Greater-Than Condition (A > B):

- A is greater than B if: (a) Most significant bit A1 > B1 (A1=1, B1=0); OR (b) A1 = B1 and A0 > B0 (A0=1, B0=0):

(A > B) = A1 . B1' + x1 . A0 . B0'.

4. Less-Than Condition (A < B):

- Similarly:

(A < B) = A1' . B1 + x1 . A0' . B0.

5. Verification:

- Exactly one output is HIGH ('1') for any combination of inputs.

>> Solution to Question 22:

Problem Statement: Explain BCD-to-7-Segment Decoders and Display Drivers (Common Anode and Common Cathode).

Problem Statement: Design a 4-to-2 Priority Encoder with truth table and K-map minimization.

driven HIGH (+5 V). Requires active-HIGH decoder (e.g., 7448).
2. Segment Truth Table for BCD Digits 0 to 9:
- Digit 0: Segments a, b, c, d, e, f active (g off).
- Digit 1: Segments b, c active.
- Digit 8: All segments a-g active.

3. Boolean Minimization (Example for Segment 'a'):
- Active for digits: 0, 2, 3, 5, 6, 7, 8, 9.
- Minimized SOP expression:

a = A + C + B.D + B'.D'.

>> [illegible] equations.

- Provides ultra-low ripple factor:
  r = sqrt(2) / [8 * omega^3 * C1 * C2 * L * RL].

[illegible]

Quine-McCluskey (Tabular) Logic Minimization:

1. Purpose:

- A rigorous algorithmic method for simplifying Boolean expressions without relying on visual inspection, making it optimal for computer-aided logic synthesis.

2. Step-by-Step Procedure:

- Step 1: Group minterms by the count of 1s in their binary representation.
- Step 2: Compare minterms between adjacent groups differing by exactly one bit position; combine them into product terms with a '-' (dash) representing the eliminated variable.
- Step 3: Repeat matching until no further reductions are possible. Unmatched terms form Prime Implicants (PIs).
- Step 4: Construct Prime Implicant Table. Rows represent PIs, columns represent original minterms.
- Step 5: Identify Essential Prime Implicants (columns with a single checkmark).
- Step 6: Select minimal remaining PIs to cover uncovered minterms.
